

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT) DEVANG PATEL INSTITUTE OF ADVANCE TECHNOLOGY & RESEARCH (DEPSTAR)		
PROJECT PROPOSAL SUBMISSION		
Semester:	5	Project ID: Indus3
Details of Project Team		
Student IDs: 24DCS057 , 24DCS029		
Sr. No:	Parameters:	
A.	Project Title	AI-Based Interview Preparation Platform
B.	Domain of Project Definition	Artificial Intelligence (AI), Web Development, Career Technology (HR Tech), Educational Technology (EdTech)
C.	Project Objectives and Scope: Project objectives are specific, measurable goals that the project aims to achieve. They provide a clear direction and serve as benchmarks for evaluating the project's success. Scope: The project scope outlines the boundaries and deliverables of the project, detailing what will and will not be included. It sets the context for what the project will accomplish and helps manage stakeholder expectations.	Objectives: The primary objective of this project is to develop an AI-powered web platform that helps students and professionals prepare for job interviews through personalized mock interviews, resume analysis, AI-generated feedback, and performance tracking. Specific objectives include: • Develop secure user authentication. • Analyze resumes and generate ATS scores. • Generate personalized interview questions. • Conduct AI-based mock interviews. • Evaluate answers using AI. • Track interview performance. • Generate downloadable reports. • Provide a categorized interview question bank. Scope: The platform allows users to: • Register and log in securely. • Upload resumes for AI analysis. • Practice HR, Technical and Behavioral interviews. • Receive AI-generated feedback and improvement suggestions. • Track interview history and performance. • Download interview reports. Future versions will include voice interviews, coding interviews and multilingual support.
D.	Background Study of Existing System The background study involves an examination of	Many existing interview preparation platforms provide only static question banks or basic mock

	the existing relevant systems or technology that the project seeks to improve or replace. This includes an analysis of its architecture, functionalities, strengths, and weaknesses.	interviews. Most lack personalization based on the user's resume, experience level and target job role. Premium AI-powered platforms are expensive and inaccessible for many students. Our platform addresses these limitations by combining AI-driven resume analysis, adaptive mock interviews, personalized feedback, ATS scoring and progress tracking into a single affordable solution.																								
E.	Methodology and Approach Methodology: The methodology section outlines the systematic procedures/method/workflow/flowchart/architecture and techniques that will be used to conduct the project. This section also describes the practical steps and strategies that will be implemented to achieve the project's goals.	Methodology: The project follows an Agile development methodology with iterative development and testing. Workflow 1. Requirement Analysis 2. UI/UX Design 3. Database Design 4. Backend API Development 5. AI Integration using Google Gemini API 6. Frontend Development 7. Testing 8. Deployment 9. Documentation																								
F.	Tentative Project Plan, Timeline and individual role The project timeline provides the project schedule, highlighting key milestones, deadlines, and the duration of each phase.	<table> <tr> <th>Phase</th><th>Timeline</th><th>Deliverable</th></tr> <tr> <td>Requirement Analysis</td><td>July 2026</td><td>Proposal, SRS</td></tr> <tr> <td>UI/UX Design</td><td>July 2026</td><td>Wireframes, User Flow, UI Mockups</td></tr> <tr> <td>Database Design</td><td>July 2026</td><td>ER Diagram, Database Schema</td></tr> <tr> <td>Backend Development</td><td>Aug 2026</td><td>FastAPI APIs, Authentication, Database Integration</td></tr> <tr> <td>Frontend Development</td><td>Aug–Sept 2026</td><td>Resume Analysis, AI Mock Interview, AI Feedback</td></tr> <tr> <td>AI Integration</td><td>Sept 2026</td><td>React.js User Interface, Dashboard, Responsive Design</td></tr> <tr> <td>Testing</td><td>Oct 2026</td><td>Functional Testing,</td></tr> </table>	Phase	Timeline	Deliverable	Requirement Analysis	July 2026	Proposal, SRS	UI/UX Design	July 2026	Wireframes, User Flow, UI Mockups	Database Design	July 2026	ER Diagram, Database Schema	Backend Development	Aug 2026	FastAPI APIs, Authentication, Database Integration	Frontend Development	Aug–Sept 2026	Resume Analysis, AI Mock Interview, AI Feedback	AI Integration	Sept 2026	React.js User Interface, Dashboard, Responsive Design	Testing	Oct 2026	Functional Testing,
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				Integration Testing, Bug Resolution
		Deployment	Oct 2026	Live Web Application on Vercel/Render
		Documentation	Oct 2026	Final Report, User Manual, Project Documentation
		Individual Responsibilities Dhruti Movaliya(24DCS057) • Project Planning & Documentation • UI/UX Design • React.js Frontend Development • Dashboard Development • Prompt Design & AI Output Testing • Testing & Report Preparation • Presentation Dhrupal Godhani(24DCS029) • Database Design • FastAPI Backend Development • Authentication & API Development • Google Gemini API Integration • AI Resume Analysis & Mock Interview Logic • Deployment & Technical Documentation		
G.	Innovation and Originality Innovation: The innovation aspect of the project refers to the novel features, methodologies, or technologies introduced to address the problem in a new and effective way. Originality: Originality emphasizes the project's uniqueness and contribution to the field. It involves the creation of new knowledge, concepts, or products that have not been previously developed.	Our project is different from traditional interview preparation platforms because it provides a personalized interview preparation experience using Artificial Intelligence. Key innovative features are: • AI-based resume analysis with ATS score. • Personalized mock interviews based on the user's job role and resume. • AI-generated feedback and improvement suggestions after every interview. • Performance dashboard to track user progress. • Downloadable interview reports. • Future support for voice interviews, coding interviews, and multiple languages. The platform helps both students and working professionals prepare for interviews in a smarter and more effective way, instead of only providing a list of interview questions.		